

Neuro-Symbolic FIR Assistance System (NSFAS)

1. Vision Statement

To develop an AI-powered decision-support system that assists police officers in drafting legally accurate, explainable, and structured FIRs by combining Natural Language Understanding with Symbolic Legal Reasoning.

The system will not replace human judgment. Instead, it will act as an intelligent legal assistant that reduces drafting effort, improves consistency, and provides explainable recommendations for applicable IPC/BNS sections.

2. Problem Statement

Current FIR drafting faces several challenges:

Manual Interpretation

Officers must manually interpret victim statements and identify applicable legal sections.

Knowledge Dependency

Accuracy often depends on the experience and legal knowledge of the officer.

Time Consumption

Complex complaints require significant analysis before drafting.

Lack of Standardization

Similar incidents may be classified differently across officers.

Explainability Gap

Existing AI text-generation systems can generate content but often fail to justify why a legal section applies.

3. Core Innovation

The project combines:

Neural AI

Used for:

- Speech Recognition
- Entity Extraction
- Event Detection
- Context Understanding

Symbolic AI

Used for:

- Legal Ontology
- Rule-based Reasoning
- Section Mapping
- Legal Explanations

This hybrid architecture forms a true **Neuro-Symbolic AI System**.

4. Primary Users

User 1: Police Officer

Responsibilities

- Review complaints
 - Verify AI suggestions
 - Edit FIR
 - Finalize FIR
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User 2: Victim / Complainant

Responsibilities

- Narrate incident
 - Provide details
 - Clarify missing information
-

User 3: Administrator

Responsibilities

- Update BNS rules

- Manage legal ontology
 - Monitor system performance
-

5. User Journey

Phase 1: Complaint Submission

Victim arrives.

Option A: Voice Narration

Example:

Someone snatched my phone while I was boarding a bus near CBS Nashik.

Option B: Text Narration

Example:

My neighbour threatened me and damaged my motorcycle.

Phase 2: Neural Understanding Layer

The system extracts:

Entities

- Victim
- Accused
- Property
- Location
- Time

Events

- Theft
- Assault
- Threat
- Fraud
- Harassment

Context

- Intent
- Violence involved?

- Property involved?
 - Financial loss?
-

Phase 3: Clarification Engine

If confidence is low, the system asks:

- Was physical force used?
- Was the accused known to you?
- Was property permanently taken?

This creates a dynamic interview process.

Phase 4: Legal Reasoning

Knowledge Graph receives facts.

Example:

```
Phone
↓
Movable Property

Taken Without Consent
↓
Theft Condition

Theft Condition
↓
Applicable Section
```

The Rule Engine performs inference.

Phase 5: Section Recommendation

Recommended Section	Confidence
BNS Section X	91%
BNS Section Y	84%

Phase 6: Explainability Layer

Example:

Section X was applied because the complaint indicates unauthorized removal of movable property without consent.

Phase 7: FIR Draft Generation

Generated Draft contains:

- Complainant Details
 - Incident Description
 - Legal Sections
 - AI Reasoning Notes
-

Phase 8: Officer Review

Officer can:

- Accept
 - Modify
 - Remove Sections
 - Add Sections
 - Edit Narrative
-

Phase 9: Final FIR Export

Supported Formats:

- PDF
 - DOCX
 - JSON
-

6. Functional Requirements

FR-1 Speech Processing

Input Languages

- Marathi
- Hindi

- English

Output

- Text Transcript
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FR-2 Complaint Understanding

Extract:

- People
 - Locations
 - Objects
 - Actions
 - Time
-

FR-3 Event Classification

Detect:

- Theft
 - Assault
 - Cyber Fraud
 - Harassment
 - Extortion
 - Property Damage
-

FR-4 Legal Ontology Engine

```
Crime
├─ Theft
├─ Assault
├─ Fraud
└─ Cybercrime
```

Mapped to:

- IPC
 - BNS
-

FR-5 Symbolic Rule Engine

Example:

```
IF
Movable Property
AND No Consent

THEN
Theft
```

FR-6 Explainability Engine

Must explain:

- Why a section was applied.
 - Which facts triggered it.
-

FR-7 FIR Draft Generator

Generate:

- Formal FIR Structure
 - Police-friendly Language
-

7. Advanced Features

These differentiate the system from typical college projects.

Feature 1: Contradiction Detection

Example:

Victim says:

He stole my bike.

Later:

I gave him the bike.

The system flags the contradiction.

Feature 2: Missing Information Detection

Example:

My wallet was stolen.

System asks:

- Approximate location?
 - Date of incident?
-

Feature 3: Legal Confidence Meter

Example:

- Theft Section → 92%
 - Robbery Section → 31%
-

Feature 4: Alternative Section Suggestions

Primary Recommendation

- Section A

Possible Alternative

- Section B
-

Feature 5: Reasoning Graph Visualization

```
Fact
↓
Rule
↓
Section
```

Feature 6: FIR Quality Checker

Checks:

- Missing Facts
 - Ambiguity
 - Incomplete Narratives
-

Feature 7: Similar Case Retrieval (Future Scope)

Example:

87 similar theft complaints found.

8. System Architecture

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graph TD; A[Victim Input] --> B[Speech-to-Text]; B --> C[NLP Engine]; C --> D[Entity Extraction]; D --> E[Event Detection]; E --> F[Knowledge Graph]; F --> G[Rule Engine]; G --> H[Legal Mapping]; H --> I[Explanation Generator]; I --> J[FIR Draft Generator]; J --> K[Officer Review];
```

9. Knowledge Graph Design

Entities

- Person
- Property
- Crime
- Location
- Evidence
- Legal Section

Relations

- Committed
 - Owns
 - Damaged
 - Stole
 - Threatened
 - OccurredAt
 - ApplicableUnder
-

10. Tech Stack

Layer	Technology
Frontend	React + Tailwind CSS
Backend	Python + FastAPI
NLP	spaCy + Transformers
Deep Learning	PyTorch
Knowledge Graph	Neo4j
Ontology	Protégé
Rule Engine	Python Rules Engine / Drools
Database	PostgreSQL
Vector Database (Future)	ChromaDB

11. Model Strategy

Plan A (Research Heavy)

Build SLM From Scratch

Pros

- Maximum Learning
- High Research Value

Cons

- Huge Dataset Requirement
- Long Training Time
- Risky for Final Year Projects

Recommendation: Not advised as primary plan.

Plan B (Recommended)

Use Existing SLMs:

- Gemma 3 (1B / 4B)
- Qwen 2.5 (3B)
- Phi-3 Mini
- SmolLM2
- Llama 3.2 (3B)

Recommended Models

- Qwen 2.5 3B
- Phi-3 Mini
- Gemma 3 4B

Fine-tune using:

- FIR Samples
 - Legal Documents
 - BNS Descriptions
-

12. Success Metrics

Metric	Target
Legal Section Accuracy	85%
FIR Draft Acceptance	80%
Explanation Correctness	90%
Response Time	< 5 seconds

13. Future Research Extensions

- Multi-agent Legal Reasoning
 - Court Charge-sheet Generation
 - Witness Statement Analysis
 - Legal Chatbot
 - RAG over Legal Corpus
 - FIR-to-Charge-sheet Automation
 - Real-time Police Station Deployment
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